ACADEMIC PROJECT DEFENSE

Road Damage Detection Using CNN

Binary classification of road images as Plain Road or Pothole



COMPUTER VISION · CONVOLUTIONAL NEURAL NETWORK

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01 · THE PROBLEM

Why automated pothole detection matters

01 Inspection does not scale

Manual road inspection is time-consuming and difficult to scale.

02 Images become predictions

The system classifies road images as Plain Road or Pothole using deep learning.

03 Assessment becomes repeatable

Goal: support faster, repeatable road-condition assessment.



A repeatable visual check can begin with one road image.

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02 · END-TO-END WORKFLOW

The pipeline turns road images into predictions

Each stage prepares the next decision, from raw dataset to saved model.



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03 · INPUT PREPARATION

Preprocessing helps the model generalize

01 Standardize the input

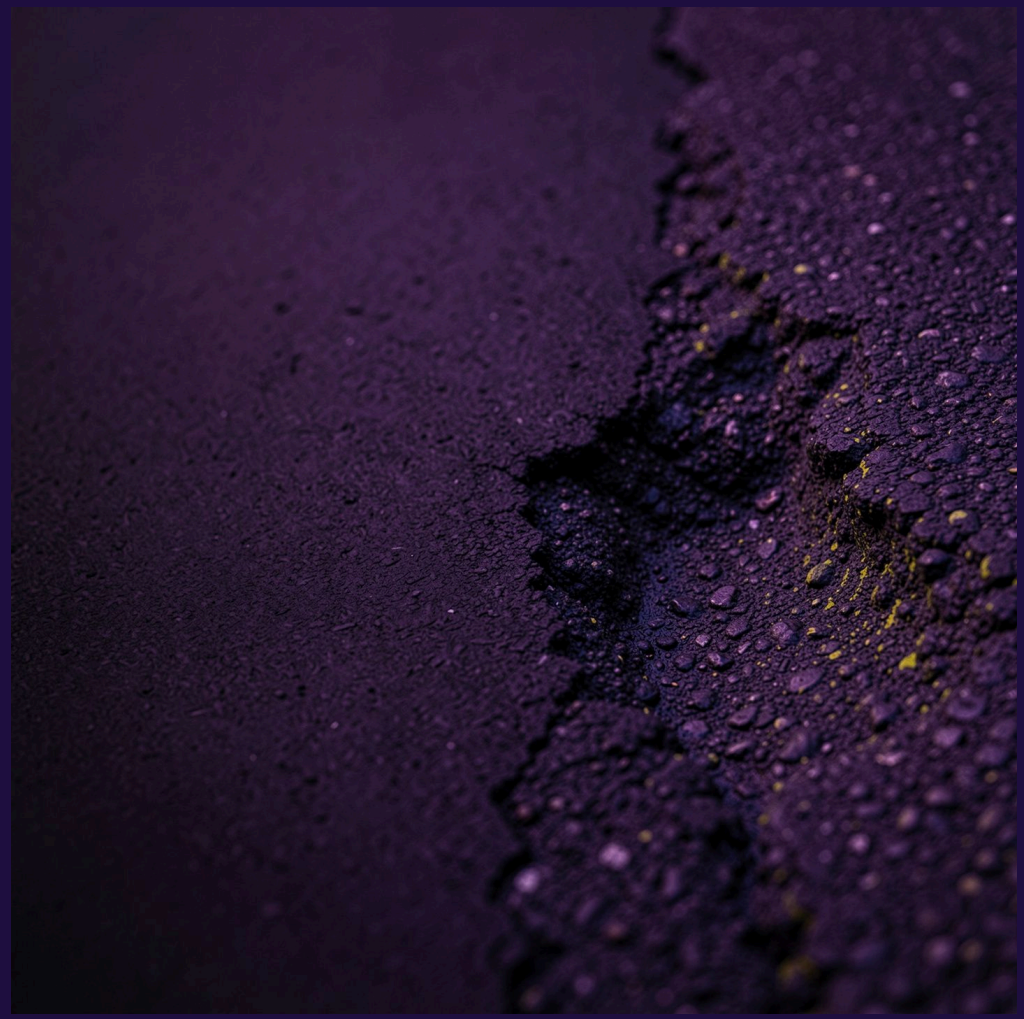
Images are resized to 224×224×3 and pixel values are scaled from 0–255 to 0–1.

02 Vary the training view

Rotation up to 15°, shifting, shearing, zooming, horizontal flipping, and nearest-neighbor filling.

03 Keep validation realistic

Validation images are only rescaled, giving a more realistic unseen-image check.



Augmentation changes the view; validation preserves the check.

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04 · MODEL ARCHITECTURE

The CNN learns from simple edges to pothole patterns

Feature extraction grows in depth; the classifier ends with one binary probability.

Feature extraction

Four convolution blocks: 32 to 256 filters, with Batch Normalization, Max Pooling, and increasing Dropout.

Global Average Pooling

Reduces spatial feature maps and parameter count.

Classifier

A 128-neuron dense layer leads to a single Sigmoid output for binary probability.

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05 · TRAINING CONTROLS

Training is designed to reduce overfitting

The training loop combines a small learning rate, regularization, and checkpoints that favor the best validation state.

0.0001

Adam learning rate

Binary Cross-Entropy loss; Accuracy metric.

50

Maximum epochs

Early Stopping restores the best weights.

BEST

Validation model

ReduceLROnPlateau adapts learning rate; ModelCheckpoint saves the best validation model.

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06 · EVALUATION

Evaluation looks beyond one accuracy number

96.53%

Earlier best validation accuracy

60-EPOCH RUN

97.58%

Earlier training accuracy

60-EPOCH RUN

Later fluctuations suggested overfitting. Final improved-model accuracy must be taken from the completed training run, not assumed in advance.

Curves

Accuracy and loss show learning behavior over time.

Class metrics

Precision, recall, and F1-score reveal class-specific errors.

Error pattern

The confusion matrix shows where predictions are wrong.

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07 · NEXT STAGE

From a strong prototype to a field-ready system

01 Current scope

Binary classification, not pothole localization with bounding boxes.

02 Next steps

Larger diverse datasets, transfer learning, object detection or segmentation, and independent real-world testing.

03 Deployment path

The trained Keras model can later support a Flask/FastAPI or mobile inspection application, with GPS-based road-damage mapping.



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